

Supporting Information

Reversible β -Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation

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1 Recombinant Synthesis and Purification of (VPAVG)₃₀

1.1 Materials

Shrimp alkaline phosphatase, PfuUltra II Fusion HS DNA polymerase, GoTaq Green Master Mix, and T4 DNA ligase with PEG4000 additive were purchased from Thermo Fisher Scientific. Exonuclease I was purchased from New England Biolabs. CircLigase ssDNA ligase was purchased from Lucigen, and KOD Hot Start DNA polymerase was purchased from Sigma-Aldrich. T4 DNA ligase and CircLigase ssDNA ligase are distinct enzymes used in separate steps of the gene-construction workflow. PCR products and plasmids were purified using Monarch PCR and DNA Cleanup kits and Monarch Plasmid Miniprep columns. DNA was recovered from agarose gels using the Monarch DNA Gel Extraction kit. Oligonucleotides and the 5'-phosphorylated synthetic template used for rolling-circle amplification were obtained from Integrated DNA Technologies. *E. coli* BL21(DE3)-Gold cells were obtained from Stratagene.

1.2 Gene construction

The repetitive ELP gene was prepared by overlap-extension rolling-circle amplification (OERCA).[1] A single-stranded DNA template encoding five VPAVG repeats was circularized in a 20 μ L reaction containing 10 pmol template, 2 μ L of 10 $\times$ CircLigase buffer, 1 μ L of 1 mmol L⁻¹ ATP, 1 μ L of 50 mmol L⁻¹ MnCl₂, and 100 U CircLigase. The reaction was incubated for 1 h at 60 °C, heat-inactivated for 10 min at 80 °C, purified, and treated with exonuclease I for 1 h at 37 °C to remove unreacted linear template.

OERCA was performed with 300 ng circular template, 1.5 μ L each of 50 μ mol L⁻¹ primers A1 and A2, 2 μ L of 70% GC dNTP mix, 2 μ L of 25 mmol L⁻¹ MgSO₄, 10 μ L of 10 $\times$ PfuUltra II buffer, 5 μ L DMSO, and 2 μ L PfuUltra II polymerase in a total volume of 100 μ L. Cycling consisted of 2 min at 95 °C; 30 cycles of 20 s at 95 °C, 20 s at 54 °C, and 90 s at 72 °C; and a final 10 min at 72 °C. Products above approximately 1.2 kb were isolated from a 0.8% agarose gel.

Ligation-independent cloning adapters were introduced by PCR using primers B1 and B2 (Figure S2), and the products were inserted into pMCSG53.[2, 3] Constructs were transformed into *E. coli* BL21(DE3)-Gold, screened for soluble expression, and verified by Sanger sequencing. The selected construct encoded the amino-acid sequence MHHHHHSSGVDLGTENLYFQSSNAV(VPAVG)₃₀, comprising an N-terminal affinity/fusion sequence followed by 30 VPAVG repeats.

1.3 Protein expression and purification

Fresh transformants were grown in Terrific Broth containing 10 mmol L⁻¹ MgSO₄ and 300 μ g mL⁻¹ ampicillin at 37 °C to an optical density of 0.6–0.8 at 600 nm. Cultures were cooled to 18 °C, induced with 0.2 mmol L⁻¹ IPTG, and grown overnight. Cells were harvested at 5000 $\times g$, resuspended in ice-cold lysis buffer (500 mmol L⁻¹ NaCl, 5 vol% glycerol, 50 mmol L⁻¹ HEPES at pH 8.0, 20 mmol L⁻¹ imidazole, and 10 mmol L⁻¹ 2-mercaptoethanol), treated with 1 mg mL⁻¹ lysozyme for 10 min, sonicated, and clarified at 29,000 $\times g$.[4]

The supernatant was applied to 5 mL IMAC Sepharose High Performance resin, washed with lysis buffer, and eluted with lysis buffer containing 500 mmol L⁻¹ imidazole. Purification fractions were monitored by SDS-PAGE. The purified protein was exchanged into 20 mmol L⁻¹ HEPES at pH 8.0, 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT and concentrated using a 10 kDa molecular-weight-cutoff centrifugal concentrator. Protein concentration was determined by direct A_{280} measurement on a NanoDrop spectrophotometer using the sequence-specific molar extinction coefficient applied during sample preparation. The final concentration was approximately 100 mg mL⁻¹; samples were stored at 4 °C, and freshly prepared protein was used for the X-ray measurements.

2 Sample Environments and X-ray Measurements

2.1 Temperature-dependent SAXS and WAXS

Two aliquots from the same (VPAVG)₃₀ stock were sealed in 2 mm quartz capillaries. The aliquot designated Measurement underwent more than 10 heating/cooling cycles across the thermal transition, whereas the Reference aliquot remained refrigerated at 4 °C. Both aliquots were subsequently measured at 10 °C and 30 °C during APS run 2021-1 at beamline 12-ID-B. The 13.3 keV beam had an approximate size of 100 $\times$ 150 μ m² and an incident flux of approximately 5 $\times$ 10¹¹ photons s⁻¹. [5, 6] SAXS was recorded with an Eiger2 9M detector at an approximately 2 m sample-to-detector distance, while WAXS was recorded with a Pilatus 300K detector at 0.42 m. The Pilatus detector was mounted above the direct-beam path, allowing the transmitted beam to continue to the SAXS detector so that SAXS and WAXS were acquired

simultaneously with a 0.1 s exposure. Buffer and capillary backgrounds were subtracted, and the SAXS and WAXS profiles were stitched over their common q interval.

2.2 Isothermal SA-XPCS

SA-XPCS was performed at APS 8-ID-I. Approximately 100 mg mL⁻¹ (VPAVG)₃₀ in 20 mmol L⁻¹ HEPES (pH 8), 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT was loaded into an aluminum cell containing a 3 mm-diameter, 2 mm-thick liquid chamber sealed by 0.127 mm polycarbonate windows (Figure S1). The cell was inserted into an anodized-aluminum holder in one temperature zone of the Quantum Northwest 3×3 stage. Each zone was driven by a Peltier element and a resistive heater, and a calibrated resistance-temperature detector (RTD) embedded adjacent to the Peltier stage supplied the controller feedback and readout.[7] The aluminum holder and aluminum cell provided direct conductive thermal coupling. Samples were held at 6 °C before measurement. A partially coherent 10.91 keV X-ray beam with 0.03% FWHM bandwidth and incident flux 1.2×10^{10} photons s⁻¹ was focused to approximately $10 \times 10 \mu\text{m}^2$. Coherent scattering was recorded 7.8 m downstream with a Rigaku XSPA-500k photon-counting detector comprising 1024×512 pixels with a 76 μm pitch and a 320 μm -thick silicon sensor. The detector was operated at its maximum 52 kHz frame rate in 2-bit mode; each acquisition contained 100,000 frames, corresponding to 19.2 μs per frame and 2 s total duration.[8]

For the isothermal experiment, the loaded sample was first held at 6 °C for more than 30 min. It was then heated to 20 °C at 5 °C min⁻¹, followed by a slower ramp from 20 °C to the target temperature of 30 °C at 1 °C min⁻¹. The two-stage protocol was used to reduce overshoot near the target temperature. The waiting time t_w was set to zero when the calibrated RTD readout reached 30 °C; data collected during the preceding ramps were not included in the isothermal waiting-time series. SAXS profiles were obtained by temporal and azimuthal averaging. Correlation functions were extracted at selected elapsed times once the low- q intensity provided sufficient photon statistics.

2.3 Experimental replication and spatial sampling

The isothermal SAXS/SA-XPCS sequence was acquired from one freshly loaded sample, and no independent sample replicate was collected. Each 2 s acquisition was recorded at a previously unexposed spatial position, providing repeated spatial sampling within the same preparation but not independent sample replication. The five q bins probe different length scales within the same time series and likewise are not independent replicates. Uncertainties shown for fitted quantities are one-standard-deviation uncertainties propagated from the correlation-function errors through the nonlinear fit; they do not represent between-sample variability.

3 SAXS/WAXS Reduction and Absolute-Intensity Calibration

3.1 12-ID-B SAXS/WAXS processing

The beamline-reduced .avg files contained q , $I(q)$, and the corresponding intensity uncertainty. For each aliquot and temperature, the capillary-plus-buffer profile was subtracted using a factor of 1.00 for SAXS and the empirical factor of 0.95 used in the original WAXS reduction; sample and background uncertainties were added in quadrature. The WAXS intensity and uncertainty were then multiplied by the ratio of the mean SAXS and WAXS intensities over their common q interval, and the two profiles were concatenated and sorted by q . The uncertainty of this overlap scaling factor was not propagated.

For Figure 2, the low- q slope was obtained by an unweighted linear fit of $\log_{10} I$ against $\log_{10} q$ over

$0.012 < q < 0.040 \text{ \AA}^{-1}$ for the Reference aliquot at 30°C . The two WAXS features were fitted independently to a Gaussian plus linear background over $0.48 < q < 1.00$ and $1.18 < q < 1.70 \text{ \AA}^{-1}$, respectively. The reduced profiles and scripts implementing the background subtraction, overlap scaling, Guinier analysis, power-law fit, and peak fits are available in the public repository identified in Section 6.

3.2 8-ID-I absolute-intensity calibration

For the 8-ID-I data, the differential scattering cross section was calculated as

$$I(q) = A_s I_s(q) - A_b I_b(q), \quad (\text{S1})$$

where $I_s(q)$ and $I_b(q)$ are the detector intensities from the sample and buffer. The conversion factors were

$$A_j = (F_j T_j t_{\text{exp}} L \Delta\Omega)^{-1}, \quad j \in \{s, b\}, \quad (\text{S2})$$

where F_j is the incident photon flux, T_j is the transmission, t_{exp} is the exposure time, L is the sample thickness, and $\Delta\Omega = (d_{\text{pix}}/D)^2$ is the solid angle per detector pixel.[9]

Sample transmission was obtained from ion chambers immediately upstream and downstream of the sample. Across the attenuation series, the transmission of the intervening air path was independent of beam intensity within uncertainty, with a mean value of 0.8679 ± 0.0009 (Figure S3a; mean $\pm$ standard deviation). The upstream ion-chamber signal was converted to incident photon flux by calibration against a PIN diode positioned in the flight path upstream of the beamstop. The measured photon flux varied linearly with the upstream ion-chamber readout (Figure S3b). The absolute-scale SAXS evolution in Figure S6 used this transmission and flux calibration.

4 SA-XPCS Correlation Analysis

The intensity autocorrelation function was calculated using a multi-tau algorithm,

$$g_2(q, \Delta t) = \frac{\langle I(q, t) I(q, t + \Delta t) \rangle_t}{\langle I(q, t) \rangle_t^2}. \quad (\text{S3})$$

Azimuthal averaging was performed in five contiguous bins of nominal width $1.13 \times 10^{-3} \text{ \AA}^{-1}$, centered at $q = 0.00376, 0.00489, 0.00602, 0.00714$, and $0.00827 \text{ \AA}^{-1}$. The centers correspond to $2\pi/q$ values of approximately 167, 128, 104, 88, and 76 nm.

The two-step correlation functions were parameterized as

$$g_2(q, \Delta t) = 1 + \beta \left[f \exp \left[- \left(\frac{\Delta t}{\tau_{\text{fast}}} \right)^{p_{\text{fast}}} \right] + (1 - f) \exp \left[- \left(\frac{\Delta t}{\tau_{\text{slow}}} \right)^{p_{\text{slow}}} \right] \right]^2. \quad (\text{S4})$$

Here, β is the speckle contrast, f is the fitted fraction of fast dynamics, $1 - f$ is the fitted fraction of slow dynamics, τ_{fast} and τ_{slow} are effective relaxation times, and p_{fast} and p_{slow} are Kohlrausch–Williams–Watts exponents. The contrast was fixed at $\beta = 0.135$, as determined from the measured short-delay contrast, and the baseline was fixed at 1. For each elapsed time, all five q bins were fitted jointly with p_{fast} and p_{slow} shared across q ; f , τ_{fast} , and τ_{slow} were allowed to vary independently with q .

Fits were performed with `scipy.optimize.least_squares` by minimizing the error-weighted residuals $[g_{2,\text{model}} - g_{2,\text{data}}]/\sigma_{g_2}$, using the pointwise g_2 uncertainties stored in the reduced HDF files. Initial values were $p_{\text{fast}} = p_{\text{slow}} = 0.5$, $\tau_{\text{fast}} = 10^{-3} \text{ s}$, $f = 0.5$, and $\tau_{\text{slow}} = 100 \text{ s}$. Bounds were $0.2 \leq p_{\text{fast}}, p_{\text{slow}} \leq 3$, $10^{-6} \leq \tau_{\text{fast}} \leq 10 \text{ s}$, $0 \leq f \leq 1$, and $1 \leq \tau_{\text{slow}} \leq 10^4 \text{ s}$. The maximum number of function evaluations

was 40,000. Global fits with reduced $\chi^2 \geq 10$ were excluded from the plotted fit-parameter trends. The covariance matrix was estimated as $(J^T J)^{-1}$ from the Jacobian of the error-weighted residuals at the solution, and reported one-standard-deviation parameter uncertainties are the square roots of its diagonal elements. Power-law relations $\tau = Aq^\gamma$ in Figure S7 were obtained by weighted linear regression of $\ln \tau$ against $\ln q$, with $\sigma_{\ln \tau} = \sigma_\tau / \tau$.

The slow component remains only partially relaxed during the 2s acquisition and frequently yields $\tau_{\text{slow}} > 2$ s. It is therefore an effective extrapolation rather than a fully sampled terminal relaxation time. Conclusions regarding arrest are based primarily on directly observed features: the decreasing amplitude of the first decay, the rising plateau, and the systematic decrease in f , equivalently the increase in the slow fraction $1 - f$. The monotonic rise of $1 - f$ indicates that a statistically increasing fraction of the scattering ensemble is dynamically arrested within the experimental window. XPCS does not determine where those regions are located, and $1 - f$ is not a direct geometric volume fraction because scattering contributions are weighted by structure and contrast. Because the measurement does not reach infinite delay, $1 - f$ is also not identified with the formal infinite-time nonergodicity factor. Figure S7 reports all auxiliary fit parameters; Figure S8 shows the measured correlation functions and fits at the four additional q values.

5 Radiation-Damage Mitigation and Controls

A Cartesian grid of 100×100 sample positions was defined in the aluminum cell, with adjacent points separated by at least one beam diameter. Every 2s acquisition was collected at a previously unexposed position, preventing cumulative dose from tracking the elapsed time of the isothermal experiment.

An independent dose-rate test was performed after the APS Upgrade using a 20 mg mL^{-1} (VPAVG)₃₀ sample. Attenuation selected incident fluxes from 2.7×10^{11} to 4.9×10^{11} photons s^{-1} , approximately 23–41 times the flux used for the main experiment. The SAXS line shapes and g_2 decays showed no systematic flux dependence (Figure S4). Because this control used a lower protein concentration and a different beam configuration than the primary experiment, it tests for gross flux-driven artifacts rather than reproducing the complete main-measurement condition.

6 Code and Reduced-Data Availability

The reduced datasets and all analysis and figure-generation code used for this manuscript are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the reduced 12-ID-B SAXS/WAXS data, averaged 8-ID-I XPCS HDF files, radiation-control data, absolute-intensity calibration inputs, and scripts used to generate the main-text and Supporting Information figures. The repository README maps each script and output file to the corresponding manuscript figure. Raw detector-frame datasets are not included in the repository and are available from the corresponding authors upon reasonable request.

7 AI-Assisted Analysis-Code Development

The Python scripts used for data processing, quantitative analysis, fitting, uncertainty propagation, and generation of data-derived figures were developed using Anthropic Claude accessed through Argo, Argonne National Laboratory’s internal generative AI interface. The scripts read the experimental and reduced SAXS/WAXS and XPCS datasets; perform background subtraction, SAXS/WAXS stitching, absolute-intensity calibration, Guinier and peak fitting, global two-mode XPCS fitting, uncertainty

propagation, and q -dependent scaling analysis; and generate the data-derived plots used in the manuscript and Supporting Information. The authors audited the scripts line by line, verified the analysis inputs and assumptions, reran the full analysis, regenerated all data-derived figures, and confirmed the reported numerical outputs. The scripts operate on measured or reduced data and do not generate experimental observations. All analysis choices, scientific interpretations, and reporting decisions were independently reviewed and approved by the authors.

8 Supporting Structural and Dynamical Results

8.1 Macroscopic reversibility

Video S1 provides a direct visual demonstration of thermal reversibility. Brief warming of a capillary containing (VPAVG)₃₀ produces turbidity within approximately 5 s, indicating structures large enough to scatter visible light. After removal of the heat source, the sample returns to optical transparency in approximately 60 s. This recovery is consistent with the overlap of the cycled Measurement and refrigerated Reference SAXS/WAXS profiles at 10 °C. Optical transparency does not specify molecular structure, but independently confirms dissolution of the mesoscale assembly on cooling.

8.2 Low-temperature unassembled state

The Reference profile at 10 °C contains a Guinier regime satisfying $q_{\max}R_g = 1.246$ (Figure S5). A linear fit of $\ln I(q)$ against q^2 gives

$$R_g = 22.65 \pm 0.50 \text{ \AA}. \quad (\text{S5})$$

The fitted size is molecular in scale and is consistent with a dispersed, unassembled low-temperature state rather than persistent mesoscale aggregates. Because the protein concentration is high, this value is reported as an apparent R_g ; weak intermolecular structure-factor contributions cannot be excluded. Recovery of the same low-temperature profile after repeated cycling establishes that the large high-temperature assembly is no longer detectable after cooling.

8.3 Isothermal SAXS evolution at 30 °C

Figure S6 shows the full SAXS sequence. At $t_w = 0$, the profile is close to the 6 °C reference. The low- q intensity then increases continuously and ultimately by more than two orders of magnitude, documenting progressive growth of mesoscale density contrast. The profile remains broad throughout, consistent with hierarchical network development rather than selection of a narrow domain spacing. The most pronounced increase occurs between $t_w = 4088$ and 5039 s, after which the intensity continues to grow through 7863 s. The correlation analysis in Figure 3 uses $t_w = 5039$ – 7863 s; earlier scattering was insufficient for reliable two-step correlations at the 19.2 μ s frame time.

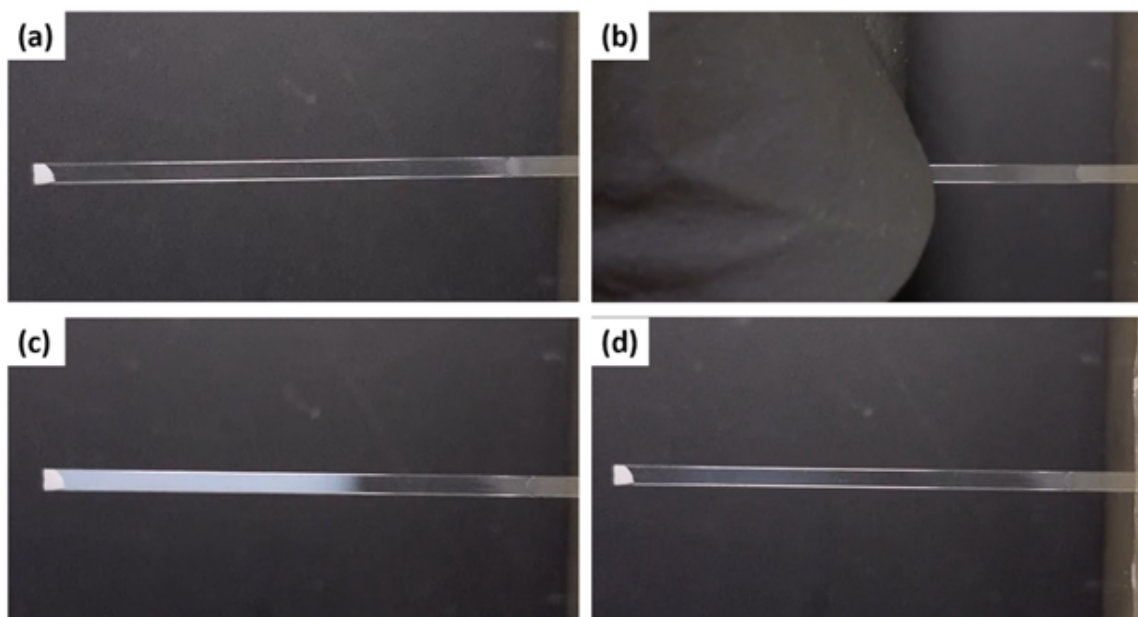
8.4 Length-scale-dependent relaxation parameters

Figure S7 shows parameters from the global fits. Both shared stretching exponents remain below unity: $p_{\text{fast}} \approx 0.5$ – 0.7 and $p_{\text{slow}} \approx 0.25$ – 0.5 . These stretched decays indicate broad relaxation-time distributions, consistent with a heterogeneous, nonequilibrium network.[10, 11]

The effective fast relaxation occurs on approximately 10^{-4} – 10^{-2} s and generally becomes faster with increasing q . At each elapsed time, the q dependence was parameterized as $\tau(q) \propto q^\gamma$. The fast-mode exponent is close to -2 over most of the analyzed interval, but becomes poorly constrained at the final

elapsed time as f approaches zero. The fitted slow component also decreases with q , but its tens-to-thousands-of-seconds values exceed the acquisition window; neither τ_{slow} nor γ_{slow} is interpreted as a directly measured terminal relaxation law. The robust arrest measure remains the monotonic decrease in the fitted fast fraction f —equivalently, the increase in the fitted slow fraction $1 - f$ —across all five q values.

Supporting Video



Video S1. Reversible thermal transition in (VPAVG)₃₀. (a) A thin quartz capillary containing (VPAVG)₃₀ is held in a fixture. (b) The capillary is warmed for approximately 5 s by contact through nitrile gloves. (c) The initially transparent solution becomes visibly turbid. (d) The sample returns to transparency approximately 60 s after removal of the heat source. The complete video can be viewed or downloaded at <https://an1.app.box.com/s/lsukldjnr45nhfrkobij sar1tqhba7jw>.

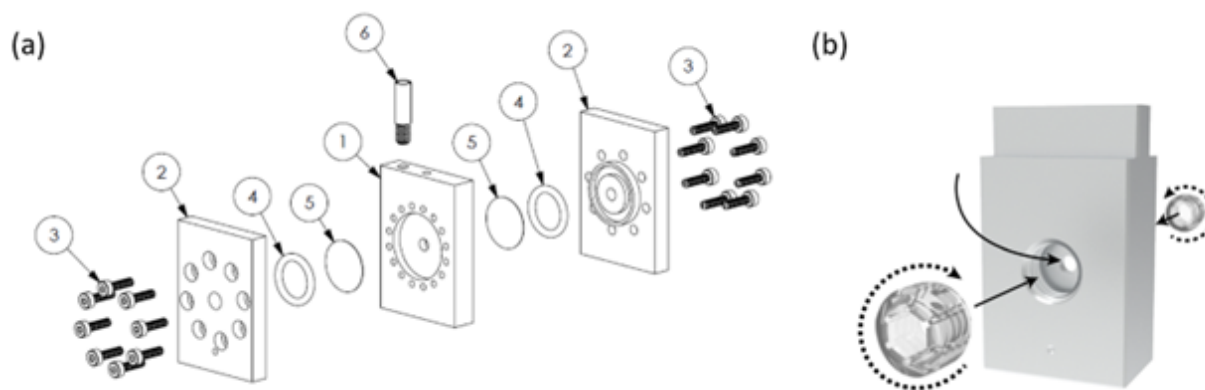


Figure S1: Sample cells used for liquid X-ray measurements. (a) Exploded view of the aluminum cell used for the 8-ID-I isothermal SA-XPCS experiment. The cylindrical chamber is 3 mm in diameter and 2 mm thick along the X-ray direction; 0.127 mm polycarbonate windows are compressed by O-rings and bolted caps. During measurement, this cell was inserted into the anodized-aluminum holder of the Peltier-controlled stage described in Section 2.2; both the holder and cell were aluminum. (b) Screw-cap cell developed separately for automated handling. Only the bolted aluminum cell in panel (a) was used for measurements reported in the main text.

```
A1 (forward)
5'-GTTCCAGCTGTTGGTG

A2 (reverse)
5'-ACCTACTGCTGGTACG

B1 (forward)
5'-TACTTCCAATCCAATGCCGTTCCAGCTGTTGGTGTGCCG

B2 (reverse)
5'-TTATCCAATTCCAATGTTAACCTACTGCTGGTACGCCGAC

VPAVG30 OERCA single-stranded template
/5Phos/GTTCCAGCTGTTGGTGTGCCGGCCGTGGGCGTGCCTGC
GGTGGGCGTCCCGGCGTCGGCGTACCAGCAGTAGGT
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Figure S2: Oligonucleotide and OERCA template sequences used to construct the ELP gene.

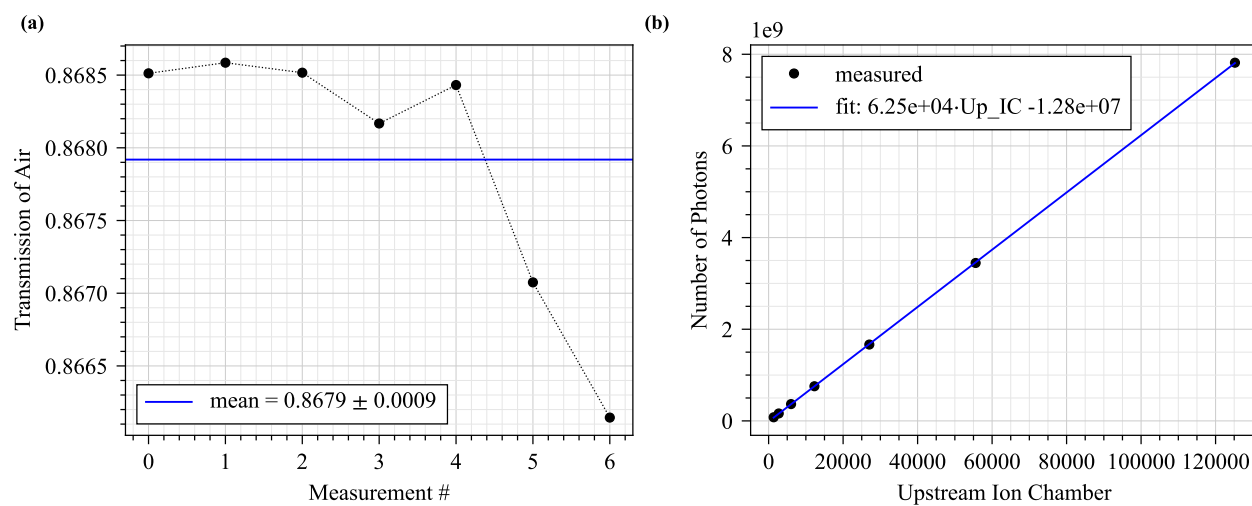


Figure S3: Transmission and photon-flux calibration for conversion to absolute scattering cross section. (a) Air-path transmission across seven measurements in the attenuation series; the line marks the mean, 0.8679 ± 0.0009 (standard deviation). (b) Photon flux measured with a calibrated PIN diode as a function of upstream ion-chamber readout; the line is the calibration fit.

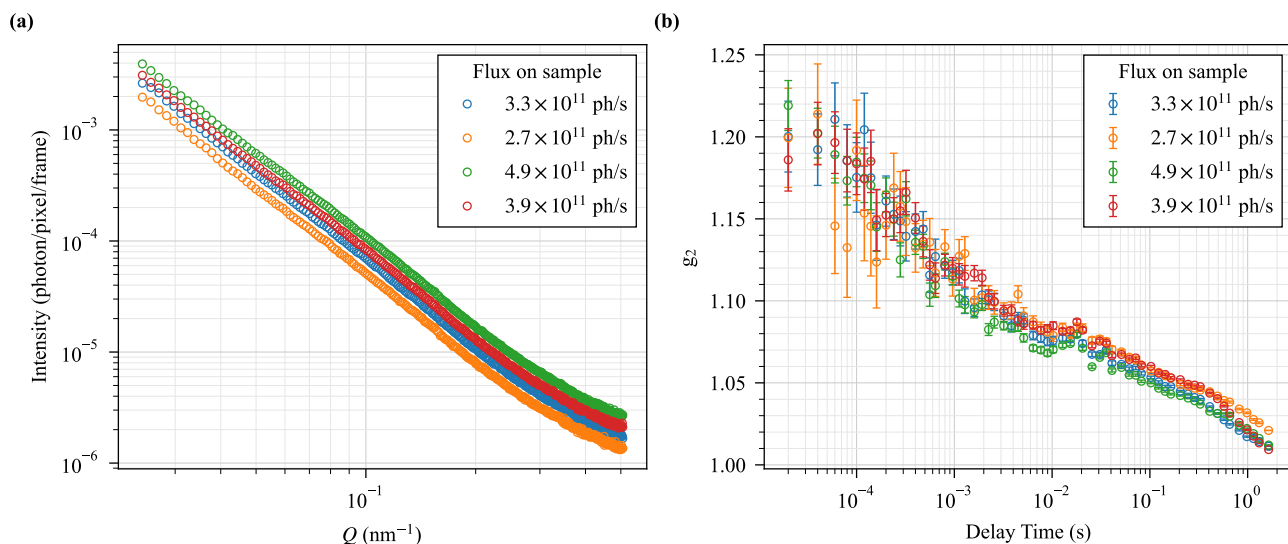


Figure S4: Flux-dependence control for SAXS and SA-XPCS. (a) SAXS profiles and (b) intensity autocorrelation functions from a 20 mg mL^{-1} (VPAVG) $_{30}$ sample at incident fluxes of 2.7×10^{11} – 4.9×10^{11} photons s^{-1} . No systematic flux-dependent change is resolved over this range, which substantially exceeds the flux used in the main experiment.

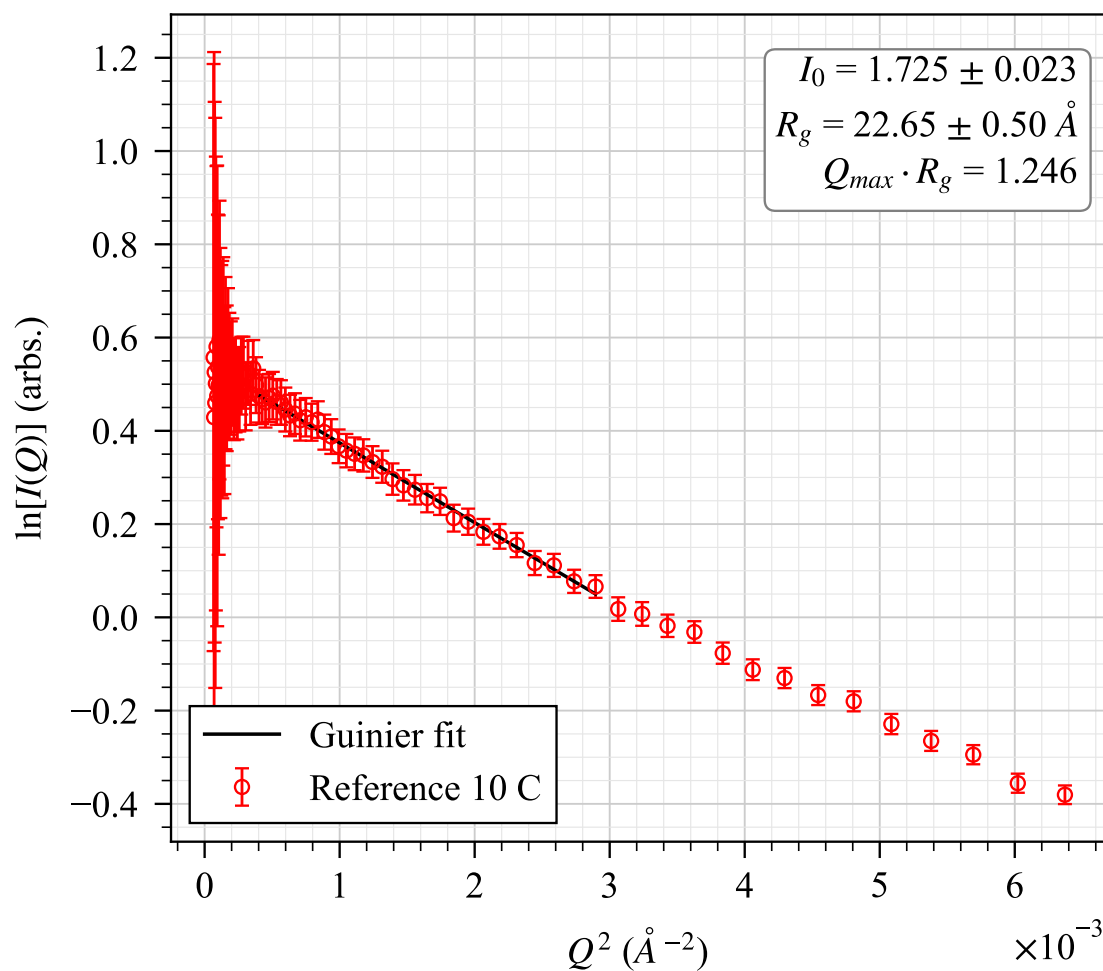


Figure S5: Guinier analysis of the Reference aliquot at 10 °C. The fit over $q_{\max}R_g = 1.246$ gives an apparent $R_g = 22.65 \pm 0.50 \text{ \AA}$ and $I_0 = 1.725 \pm 0.023$.

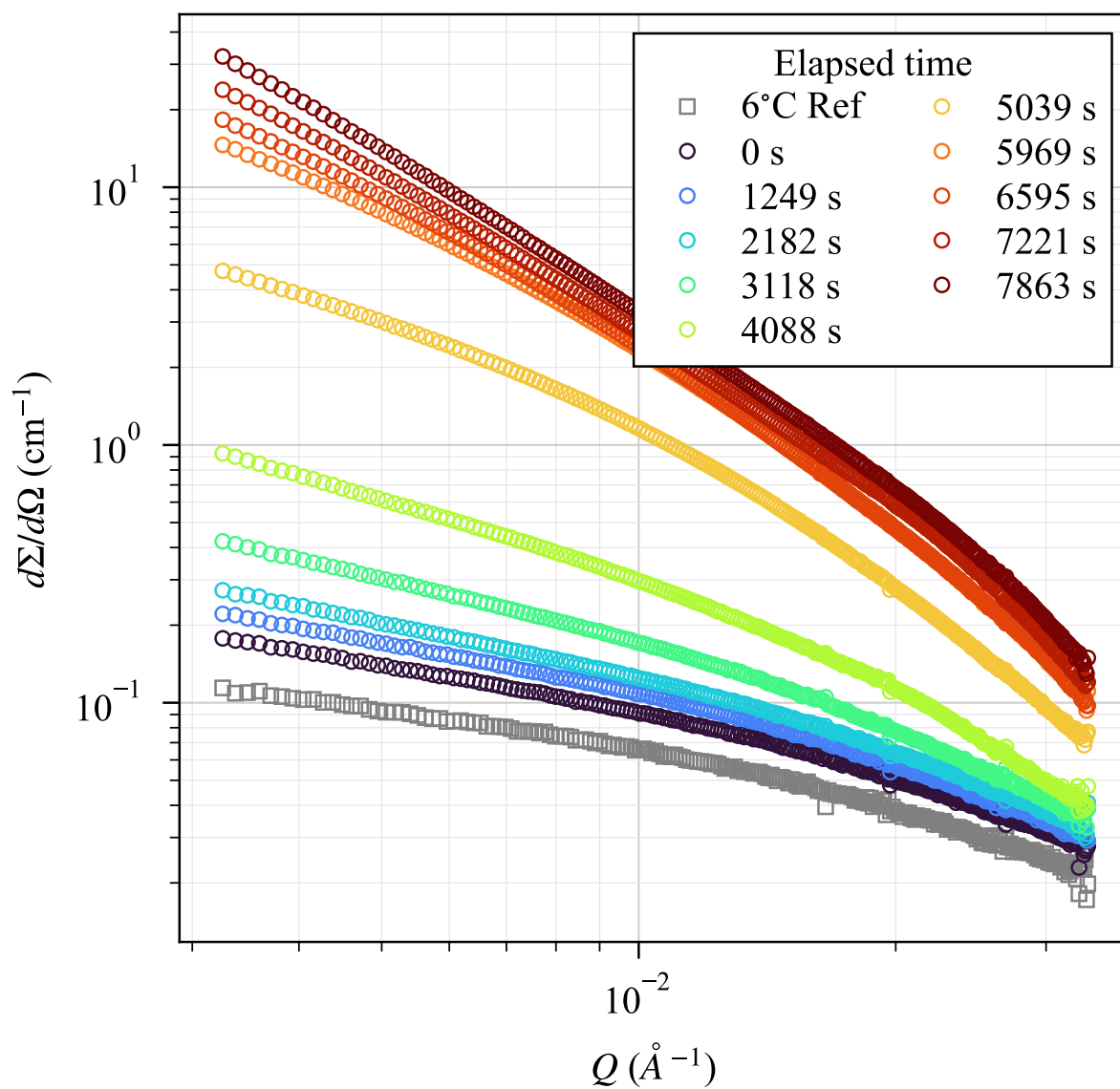


Figure S6: Complete isothermal SAXS evolution of (VPAVG)₃₀ at 30 °C. The 6 °C reference and profile at the start of the hold are followed by profiles from 1249 to 7863 s. The subset from 5039 to 7863 s corresponds to the SAXS and SA-XPCS data analyzed in Figure 3.

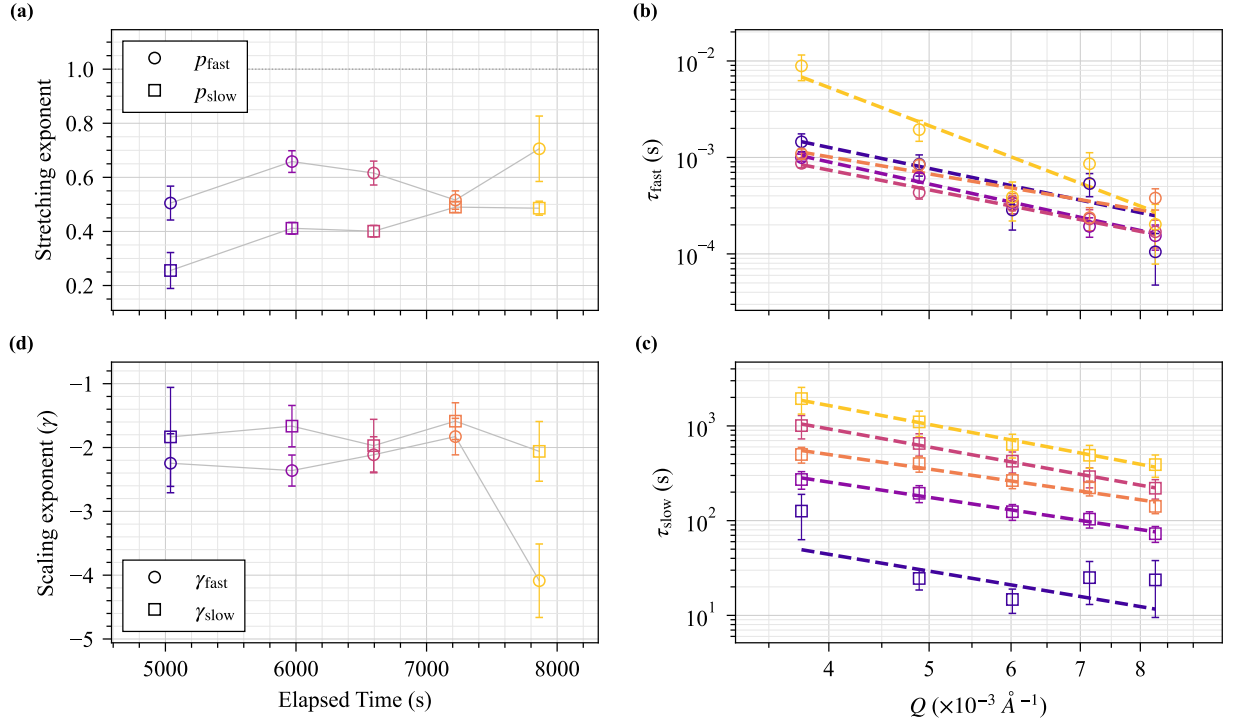


Figure S7: Additional parameters from global two-mode fits. (a) Stretching exponents shared across the five q bins at each elapsed time. (b) Effective fast relaxation time versus q . (c) Effective slow relaxation time versus q . Dashed lines in panels (b,c) are power-law fits $\tau \propto q^\gamma$. (d) Fitted scaling exponents γ_{fast} and γ_{slow} . The final fast-mode exponent is poorly constrained because f approaches zero, and the slow-mode quantities are extrapolations beyond the 2 s window. Error bars are fit uncertainties.

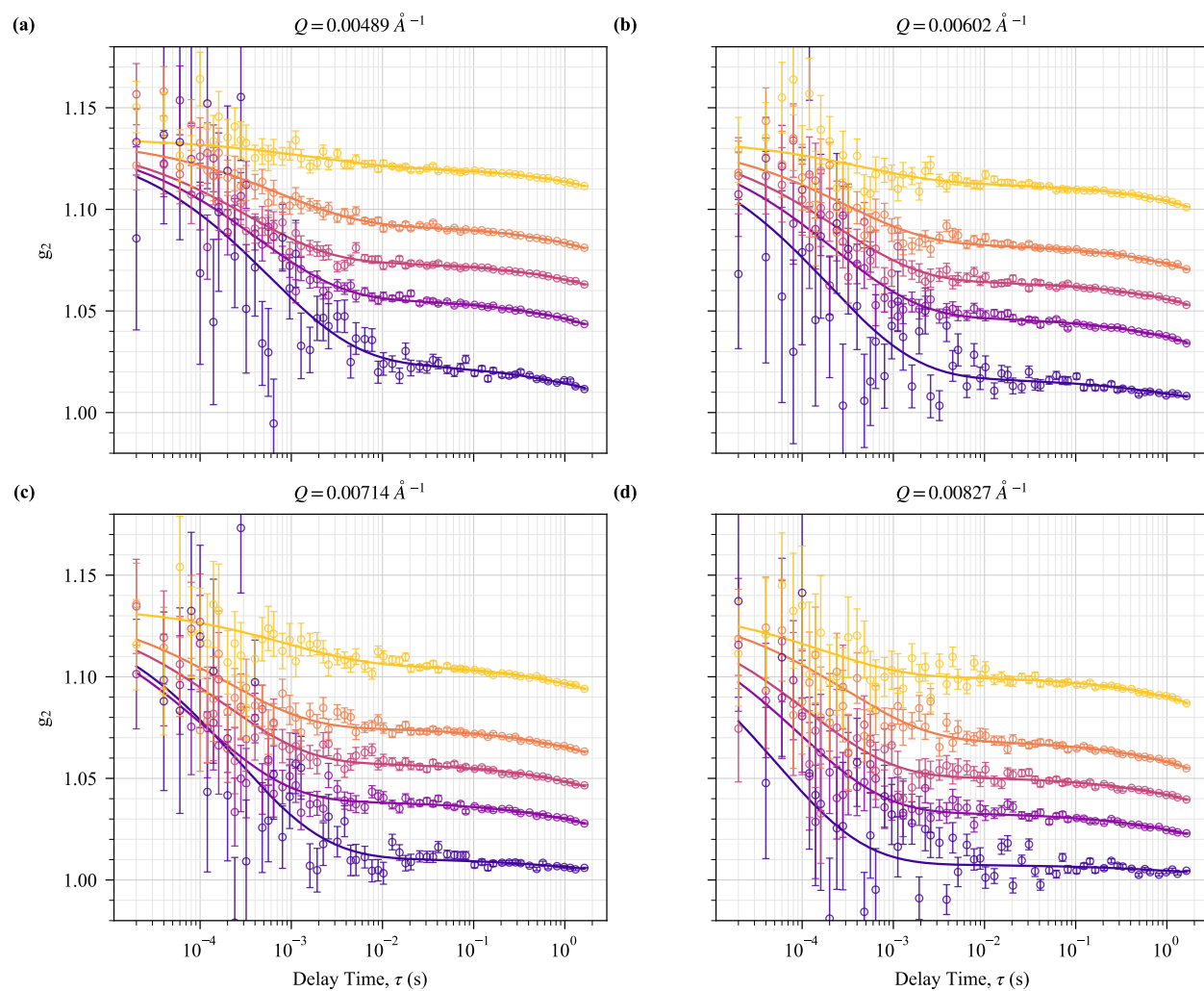


Figure S8: Correlation functions and global two-mode fits at the four additional q values. Panels show $q = 0.00489$, 0.00602 , 0.00714 , and $0.00827 \text{ \AA}^{-1}$. Colors denote elapsed times used in Figure 3; solid curves are global fits with shared stretching exponents at each elapsed time.

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